Howard University
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EECE 306-01: Fundamentals of Electromagnetics Laboratory

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Lab 9: Current Sources and the Biot-Savart Law

Objectives

By the end of this session you will have

- extended the toolbox to magnetostatics with four new functions and no change to the field evaluation loop,
- found out whether your Lab 3 and Lab 4 code was written as generally as the API required,
- validated the magnetic field engine against three closed form limits,
- reused the symmetry test pattern of Lab 3 in a magnetic setting.

1. The same loop with a different numerator

In Lab 3 the source struct was defined as a discretized list of elements, and `em.field.E` was written to loop over that list without knowing what kind of source produced it. In Lab 4 the quadrature engine was built to turn any parametrized curve, surface or volume into such a list. Both decisions were made for electrostatics. Today they are applied to magnetostatics without change.

The two kernels, side by side, are

$$\vec{\boldsymbol{E}}(\vec{\boldsymbol{r}}) = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{M} q_i w_i \frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i|^3} \qquad \vec{\boldsymbol{H}}(\vec{\boldsymbol{r}}) = \frac{1}{4\pi}\sum_{i=1}^{M} \frac{(I\,\mathrm{d}\vec{\boldsymbol{L}})_i \times (\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i)}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i|^3}$$

Both are one loop over the elements, both use the displacement vector $\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i$, and both divide by its cube. The numerator changes from a scalar times a vector to a cross product, and the constant changes from $1/(4\pi\varepsilon_0)$ to $1/(4\pi)$. As far as your code is concerned, that is the entire difference between electrostatics and magnetostatics.

How long em.field.H takes, and why the time is recorded

Writing `em.field.H` after `em.field.E` should take about fifteen minutes.
If it takes an hour, `em.field.E` was written with assumptions that the physics did not require. The two most common are that the element strength is a scalar, and that the Coulomb constant is mixed into the loop over observation points instead of being applied once at the end.
Record in your notebook how long `em.field.H` took and what had to change in your Lab 3 code. The report is graded on the diagnosis, not on the time. A team that reports two hours and explains what was refactored receives more credit than a team that reports fifteen minutes and nothing else.

2. Part I. Current sources

2.1 `em.src.lineCurrent(I, curve, tspan, N, rule)`

The element quantity is a **vector**, obtained from the curve tangent,

$$(I\,\mathrm{d}\vec{\boldsymbol{L}})_i = I\,\left.\frac{\mathrm{d}\vec{\boldsymbol{r}}}{\mathrm{d}t}\right|_{t_i} w_i^{\mathrm{node}}$$

The charge case differs here, since the element strength q_i is a scalar. A charge has no direction and a current has one. The difference is physics, not an inconsistency in the design, and the struct reflects it with `Idl` being `Mx3` while `q` is `Mx1`.

This constructor needs the tangent $\mathrm{d}\vec{\boldsymbol{r}}/\,\mathrm{d}t$ from `em.quad.line` as a vector, not only its magnitude. If your Lab 4 implementation kept only $|\,\mathrm{d}\vec{\boldsymbol{r}}/\,\mathrm{d}t|$ inside the weight and discarded the direction, `em.quad.line` needs a small change so that it also returns the tangent. Make the change there, instead of differentiating the parametrization a second time inside `em.src.lineCurrent`, so that the derivative exists in one place only.

2.2 `em.src.surfCurrent(K, surf, uspan, vspan, Nu, Nv, rule)`

`K` is a function handle returning a `1x3` surface current density in A/m at each position. The element vector is $\vec{\boldsymbol{K}}$ at the element position times the scalar area weight.

2.3 `em.src.merge` needs no change

The merge function written in Lab 3 composes current sources without modification, because it only reads the `type` field and concatenates the element arrays. Confirm that it still refuses to merge a charge source with a current source.

3. Part II. `em.field.H` and `em.field.B`

Implement the Biot-Savart sum above. `em.field.B(s, r, mur)` returns $\mu_r\mu_0$ times `H`, with `mur` defaulting to 1.

The same three requirements as for `em.field.E` apply. Vectorize over 10^4 observation points. Guard the singularity when an observation point coincides with an element position. Call `em.const.mu0()` instead of typing the number.

4. Part III. Three closed form limits and one symmetry

4.1 Circular loop, on axis

A loop of radius a carrying current I, lying in the xy plane and centered at the origin, produces on its axis

$$H_z(z) = \frac{I a^2}{2\,(a^2 + z^2)^{3/2}}$$

Compare at ten values of z with a relative tolerance. At the center, $z = 0$, the expression reduces to $I/(2a)$.

4.2 Straight segment approaching the infinite wire

A finite straight segment of length $2h$ on the z axis, evaluated at radial distance ρ from its midpoint, approaches

$$H_\phi = \frac{I}{2\pi\rho}$$

as h/ρ grows.

Plot the ratio of the computed field to this limit against h/ρ and report the value of h/ρ at which the ratio reaches 0.99.

Lab 4 contained the same study for the finite line charge approaching the infinite line charge. Compare the two threshold aspect ratios. They should agree, because both fields fall off as the inverse square of the distance to the source element, so the far part of the segment stops contributing at the same aspect ratio in both cases. Write one sentence on this in your notebook.

4.3 Solenoid, and the long limit

A solenoid of N turns, height h and radius a has, in the long limit, a uniform interior field

$$H_z = \frac{NI}{h}$$

Build it as a stack of N loops, or as a helix, with `em.src.merge`, and compare the center field with this limit. Sweep h/a and report the value at which the center field reaches 99 percent of NI/h.

Then plot H_z along the full axis from well below the solenoid to well above it. The field is not uniform near the ends. At the mouth of a long solenoid it falls to one half of the central value, and seeing that in your own data is more convincing than reading it.

4.4 The symmetry test

On the axis of the circular loop the transverse components H_x and H_y vanish by symmetry. Assert this with an absolute tolerance, justified the same way as in Lab 3.

This is the Lab 3 test pattern applied to a different field, and that is the reason it is here. A symmetry argument does not depend on which field is computed. Once the habit is formed, every new problem arrives with a test that needs no reference solution.

Required tests

Write `tests/test_lab09.m` containing at least these checks.

1. Circular loop on-axis field matches the closed form at ten values of z, relative tolerance 10^{-6}.
2. Center of the loop gives $I/(2a)$.
3. Transverse components vanish on the loop axis, absolute tolerance justified.
4. Finite segment ratio exceeds 0.99 at your reported h/ρ.
5. Solenoid center field reaches 99 percent of NI/h at your reported h/a.
6. Reversing the direction of the current reverses the field, checked componentwise.
7. Doubling the current doubles the field, confirming linearity.
8. `em.src.merge` of two current sources works, and merging a current source with a charge source raises an error.
9. `em.field.H` returns `Nx3` for $N = 5000$ in under two seconds.
10. The full suite from Lab 1 through Lab 9 passes in one run.

Deliverables

Submit through Canvas by the start of the next lab session.

1. **Toolbox snapshot**, `Team`*NN*`_Lab09.zip`, now at 47 of 53 functions.

2. **Lab notebook**, published script exported to PDF, three to five pages, with your account of how long `em.field.H` took and what was refactored, the two aspect ratio studies with the Lab 4 comparison, and the axial profile of the solenoid.
3. **Test output** pasted into the notebook.

Looking ahead

Next week only two new functions are due. With them, two of Maxwell's four equations will be verified numerically, using the integrators written in Labs 5 and 6 without modification. Read Sections 3.3 through 3.5 of Wentworth before you come.